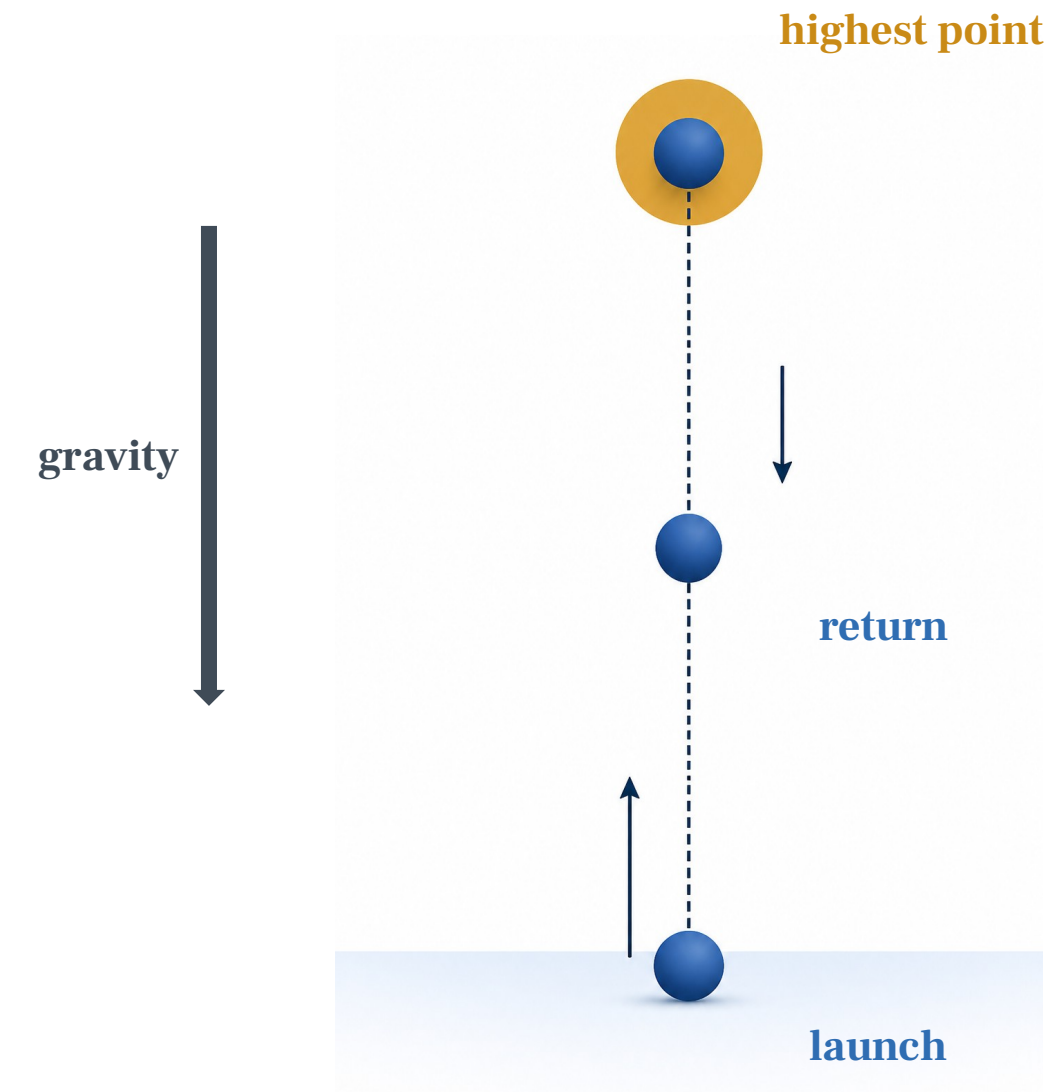


Physics to Arrays and Plots

Week 1 | From Motion to MATLAB

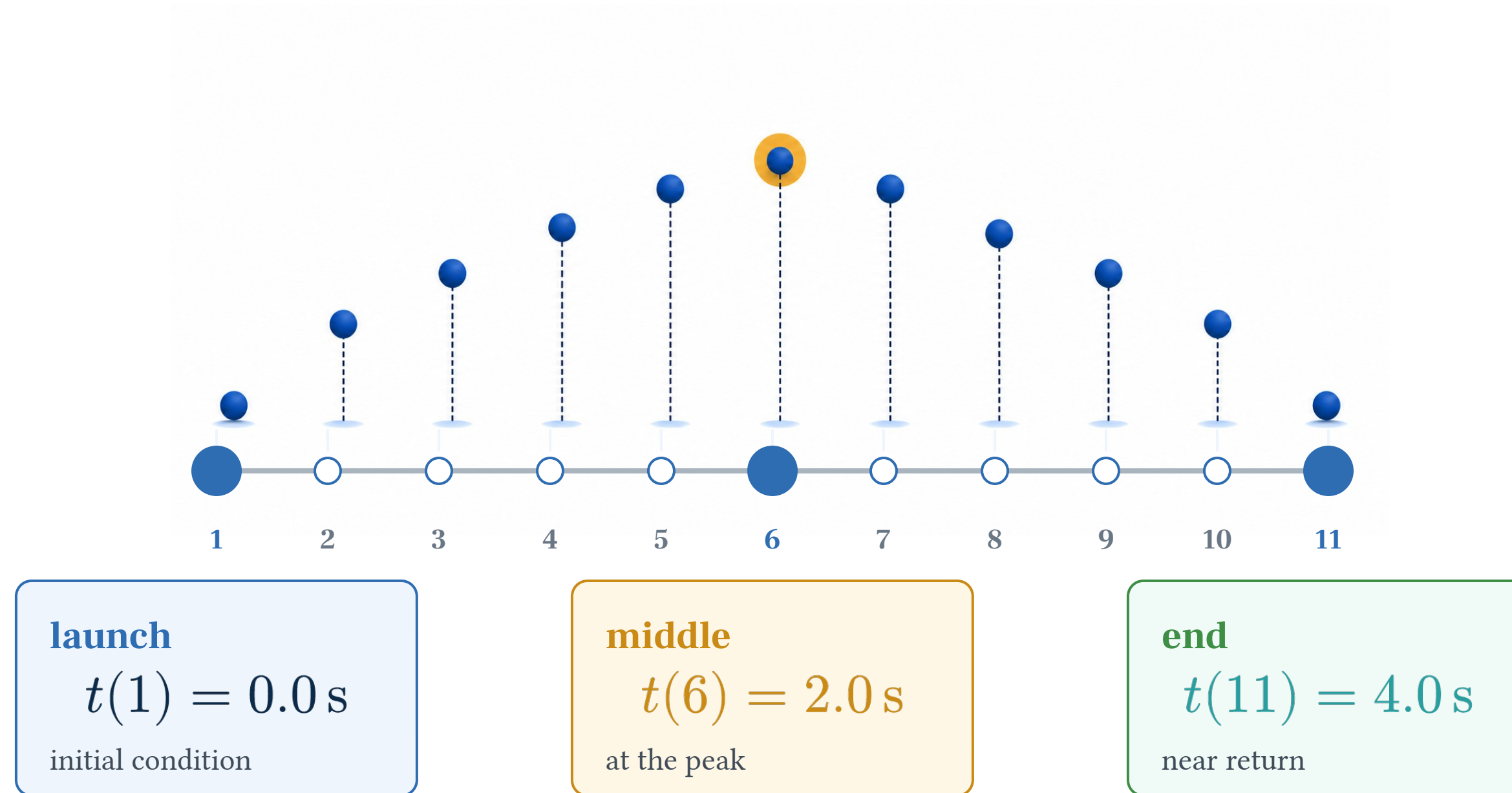


Editable PowerPoint master proof



Arrays store physical moments

Each index identifies one measurement time



The number in parentheses selects one physical moment from the array

A labelled plot turns values into evidence

Axes, units, and shape must agree with the model

Before plotting

starts at

$$y = 0 \text{ m}$$

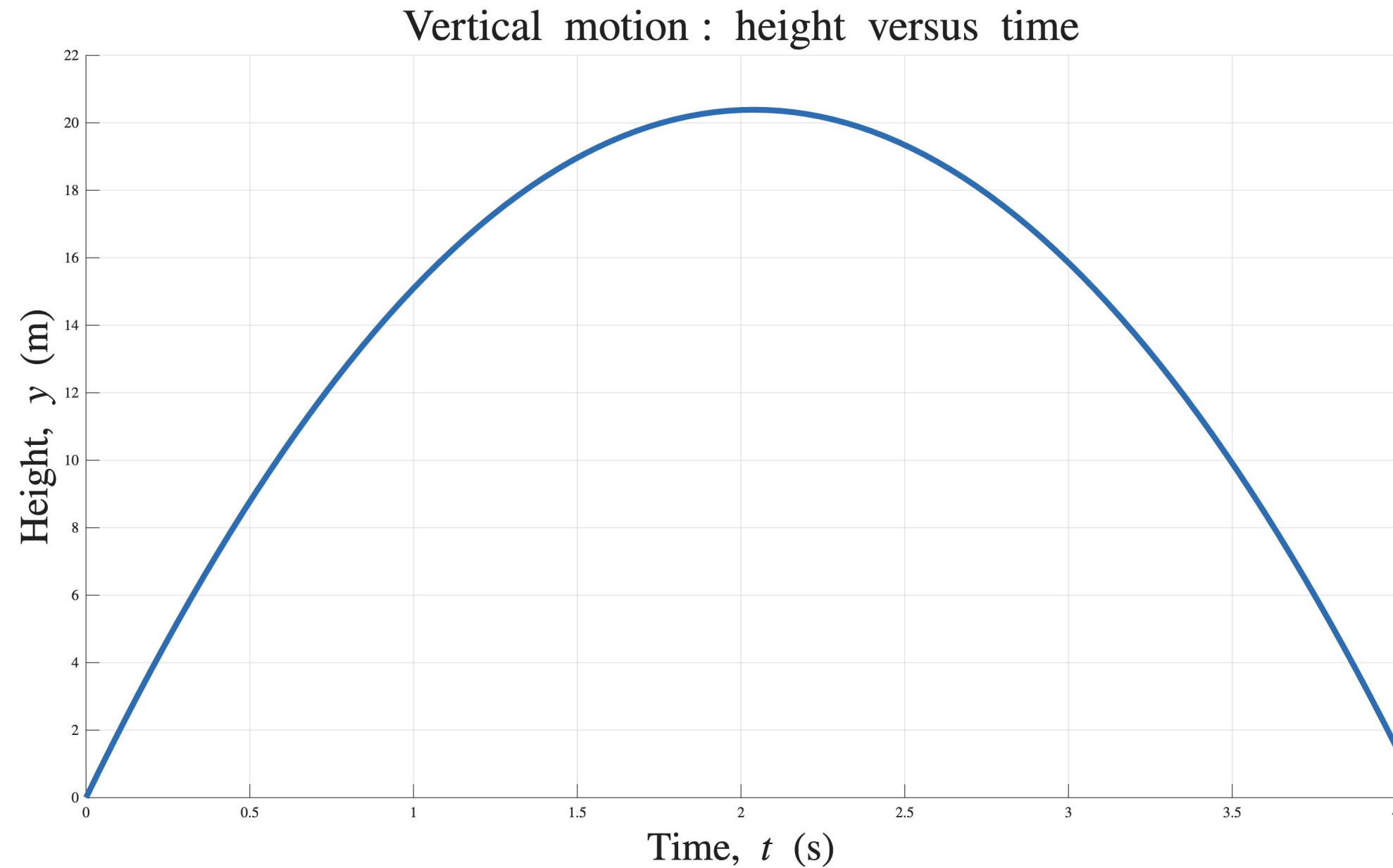
rises to

one maximum

returns to

$$y \approx 0 \text{ m}$$

prediction first

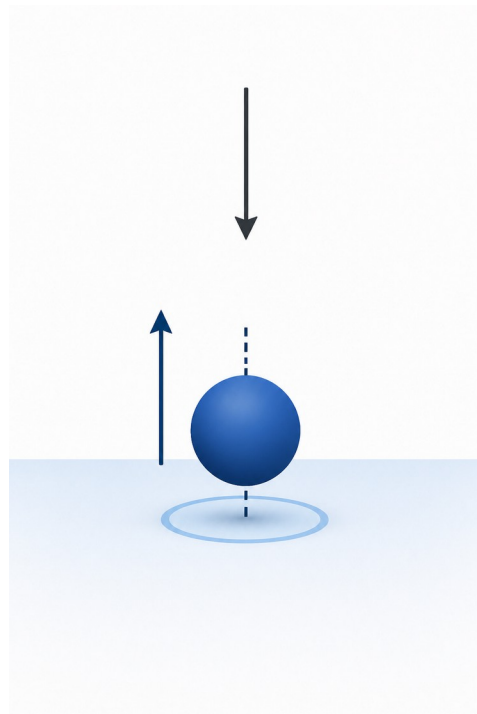


Does the curve match all three predictions?

A boundary check can reveal an error that a smooth curve hides

Use a known condition before trusting the full result

Known condition



At launch

$$t = 0 \text{ s}$$

must give

$$y = 0 \text{ m}$$

Computed result

$$y(1) = 0.000 \text{ m}$$



The first array value satisfies the launch condition

A smooth graph is not enough: test at least one value that physics already tells you

Names and units keep the model readable

Translate each physical quantity before writing code

physics symbol	meaning	MATLAB name	unit
t	time	<code>t</code>	s
y	vertical position	<code>y</code>	m
v_0	initial velocity	<code>v0</code>	m s^{-1}
g	gravitational acceleration	<code>g</code>	m s^{-2}

Readable names reduce syntax mistakes and make unit checks easier

Short code should expose the physics

The algorithm appears before the MATLAB syntax

Plain-language algorithm

- 1 Choose the model parameters
- ↓
- 2 Create an array of times
- ↓
- 3 Calculate height at every time
- ↓
- 4 Plot height against time
- ↓
- 5 Check the launch value

MATLAB

```
% model parameters
v0 = 20;    % m/s
g = 9.81;   % m/s^2

% array of times
t = linspace(0,4,11);

% height at every time
y = v0.*t - 0.5*g.*t.^2;

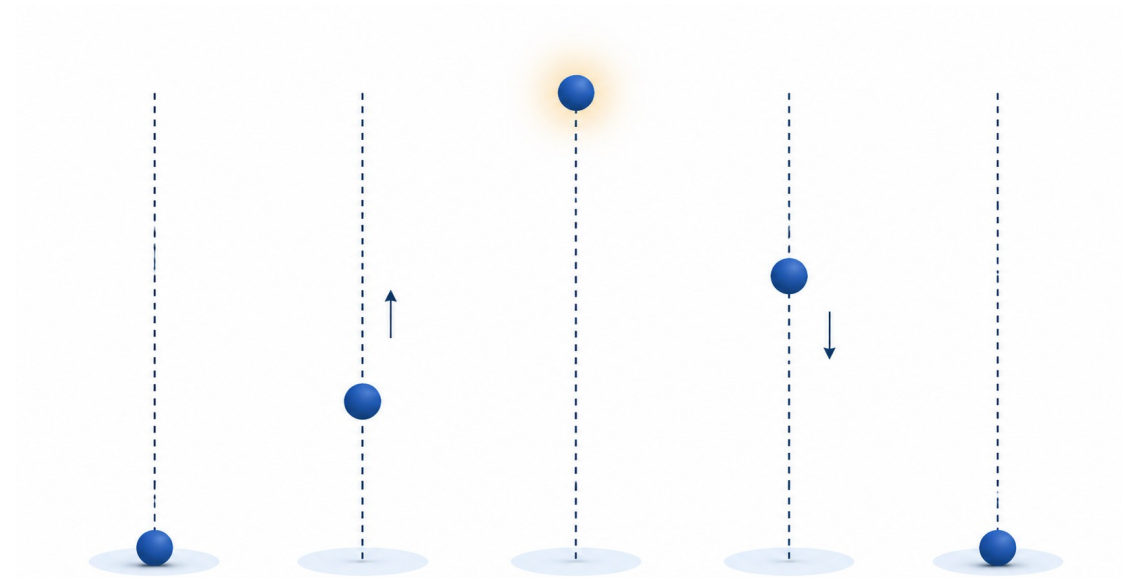
plot(t,y,'LineWidth',2)
y(1)      % boundary check
```

.* and **.^** apply the operation to every array element

The equation links each term to motion

Read the model from left to right before evaluating it

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

 y_0

starting position

 $v_0 t$

upward contribution

 $-\frac{1}{2} g t^2$

gravity contribution

 $y(t)$

predicted position

Unit check: every term on the right must have units of metres

Changing one parameter reveals the trend

Keep the model fixed and change only the initial velocity

controlled comparison

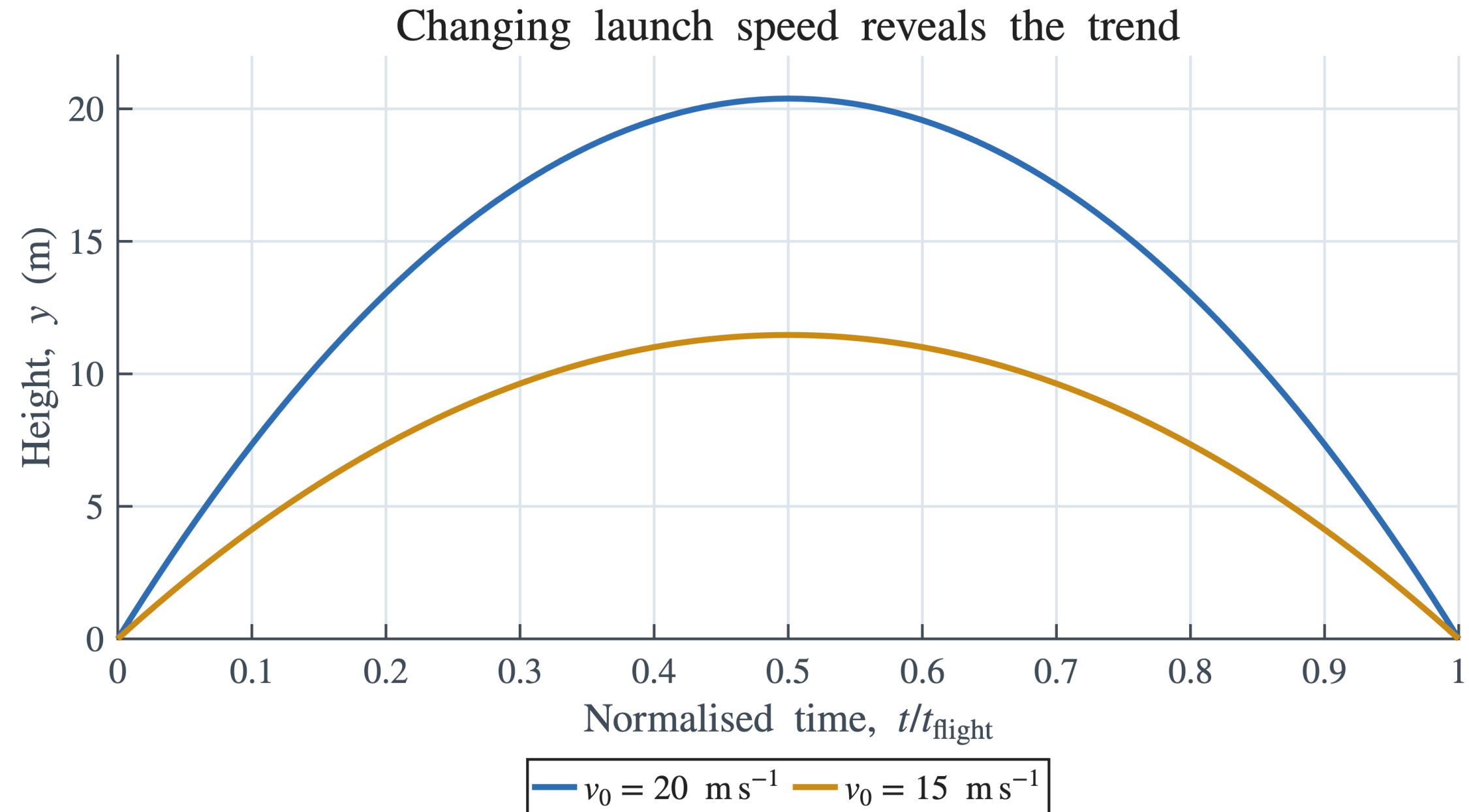
Ask one focused question

What changes when launch speed changes?

Keep gravity and starting height fixed.

Higher initial speed → higher peak

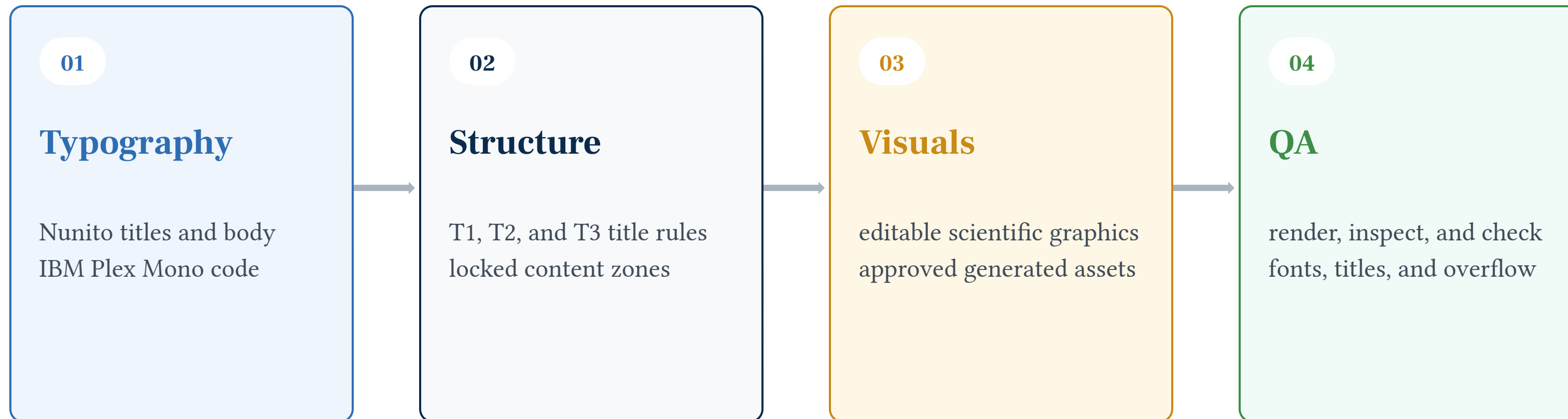
changed: v_0 fixed: g, y_0



Observed: the blue curve reaches the higher peak

The design system keeps every week coherent

Approve this proof before the Week 1 teaching deck is rebuilt



Review focus: title character, spacing, component shapes, chart treatment, and visual balance